

MAT 310, Handout 22: Multinomial Distribution (5.9)

Recall: $Y \sim \text{Bin}(n, p)$ models the number of successes in n independent trials, each with only two potential outcomes.

Multinomial Experiment

1. Fixed number (n) of **independent** trials
2. Each trial has k potential outcomes
3. p_i = probability any trial results in outcome i , so $\sum_{i=1}^k p_i = 1$
4. The RV $Y_i = \#$ of trials resulting in outcome i
5. $\sum_{i=1}^k Y_i =$

Multinomial Distribution: For $(Y_1, Y_2, \dots, Y_k) \sim \text{Multinomial}(n, p_1, p_2, \dots, p_k)$,

$$p(y_1, y_2, \dots, y_k) =$$

Expected Value, Variance, and Covariance

$$E(Y_i) =$$

$$V(Y_i) =$$

$$\text{Cov}(Y_i, Y_j) =$$

Example 1. A soccer team wins each game with probability 0.5, loses with probability 0.3, and otherwise ties. Their season consists of 20 games.

(a) Find the probability they win 12, lose 6, and tie 2 games.

(b) Teams earn 3 points for a win and 1 point for a tie. Find the expected value and variance for the number of points they earn by the end of the season.